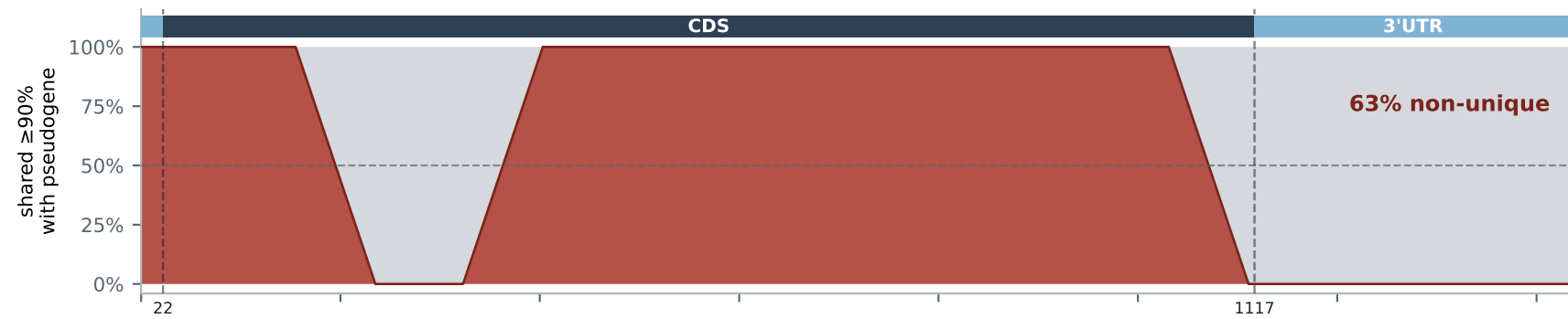
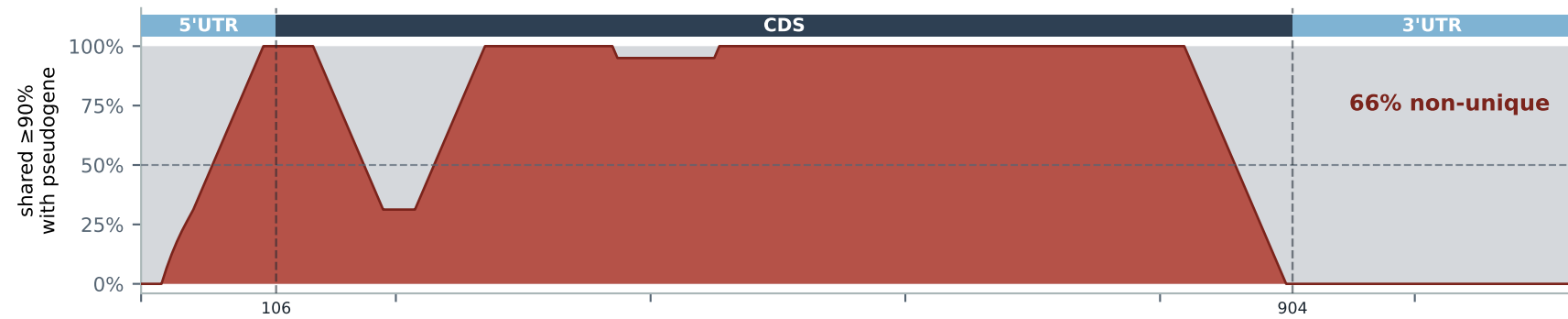


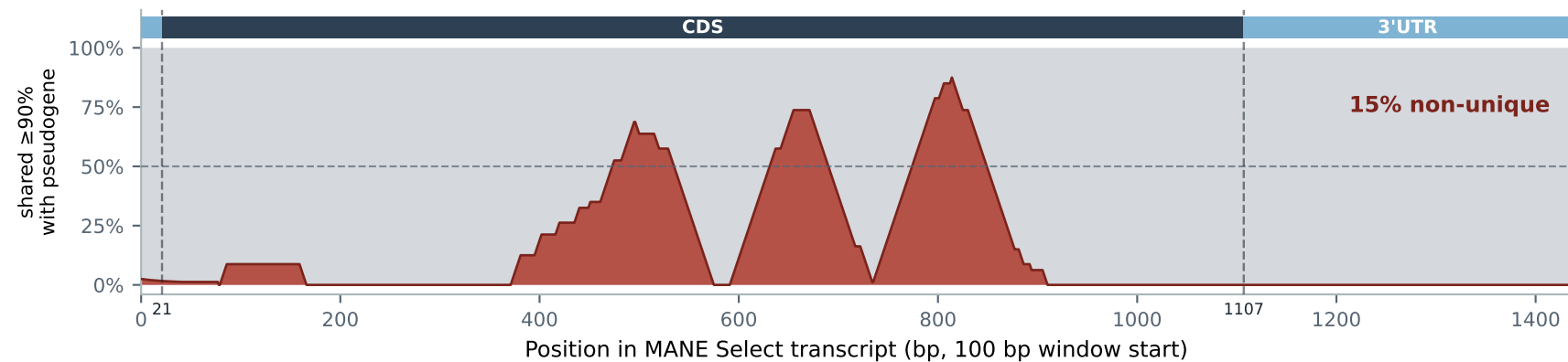
HLA-A (pseudogenes: H, J, L, P, V)



HLA-DRB1 (pseudogenes: DRB2, DRB6, DRB7, DRB8, DRB9)



HLA-B (pseudogenes: K, N, T, U, W)



$\geq 90\%$ identical to ≥ 1 pseudogene (100 bp window) unique region CDS 5'/3' UTR ---- 50%

Smoothed with 80 bp rolling mean. GENCODE v49, MAFFT L-INS-i. Fractions are lower bounds — pseudogene transcript models are partial.